DS200 – BÀI THỰC HÀNH 2

Bài 1:

```
cd "/mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy"
pig -x local Bai1.pig
```

```
dy initialized!
2026-04-10 02:14:13,739 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:14:13,746 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 10
0% complete
2026-04-10 02:14:13,750 [main] INFO  org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  thien592005     2026-04-10 02:14:08     2026-04-10 02:14:13     REPLICATED_JOIN,FILTER

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime  A
vgReduceTime    MedianReducetime        Alias   Feature Outputs
job_local1608320523_0001        1       0       n/a     n/a     n/a     n/a     0       0       0       0       stopword
s       MAP_ONLY
job_local937686203_0002 1       0       n/a     n/a     n/a     n/a     0       0       0       0       filtered_words,f
inal_result,joined_data,lower_data,raw_data,words_data  REPLICATED_JOIN,MAP_ONLY        file:///mnt/c/Users/Admin/Docume
nts/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/KetQuaBai1,

Input(s):
Successfully read 131 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/stopwords.
txt"
Successfully read 11770 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/hotel-re
view.csv"

Output(s):
Successfully stored 181204 records in: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/KetQuaB
ai1"
```

```
Counters:
Total records written : 181204
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1608320523_0001        ->      job_local937686203_0002,
job_local937686203_0002


2026-04-10 02:14:13,751 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:14:13,753 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:14:13,754 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:14:13,761 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:14:13,763 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:14:13,764 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:14:13,769 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Su
ccess!
2026-04-10 02:14:13,789 [main] INFO  org.apache.pig.Main - Pig script completed in 8 seconds and 486 milliseconds (8486
ms)
thien592005@Thien-dz:/mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy$
```

Bài 2:

```
pig -x local Bai2.pig
```

```
HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  thien592005     2026-04-10 02:19:46     2026-04-10 02:17:58     GROUP_BY,FILTER

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime  A
vgReduceTime    MedianReducetime        Alias   Feature Outputs
job_local1458515235_0001        1       1       n/a     n/a     n/a     n/a     n/a     n/a     n/a     n/a     aspect_c
ounts,category_counts,data,group_aspect,group_category  MULTI_QUERY,COMBINER    file:///mnt/c/Users/Admin/Documents/Uni/
DS200/ThucHanh/BTTH2/Bai1 - Copy/ThongKe_Category,file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Co
py/ThongKe_Aspect,
job_local1752170387_0002        1       1       n/a     n/a     n/a     n/a     n/a     n/a     n/a     n/a     cleaned_
words,group_words,high_freq_words,word_counts   GROUP_BY,COMBINER       file:///mnt/c/Users/Admin/Documents/Uni/DS200/Th
ucHanh/BTTH2/Bai1 - Copy/ThongKe_TuTren500,

Input(s):
Successfully read 11770 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/hotel-re
view.csv"
Successfully read 181204 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/KetQuaB
ai1"

Output(s):
Successfully stored 7173 records (787508 bytes) in: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 -
 Copy/ThongKe_Category"
Successfully stored 8 records (123 bytes) in: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/
ThongKe_Aspect"
Successfully stored 74 records in: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/ThongKe_TuT
ren500"
```

```
Counters:
Total records written : 7255
Total bytes written : 787631
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1458515235_0001
job_local1752170387_0002


2026-04-10 02:17:58,389 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:17:58,392 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:17:58,396 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:17:58,414 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:17:58,429 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:17:58,434 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:17:58,446 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Su
ccess!
2026-04-10 02:17:58,515 [main] INFO  org.apache.pig.Main - Pig script completed in 14 seconds and 703 milliseconds (1470
3 ms)
thien592005@Thien-dz:/mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy$
```

Bài 3:

```
pig -x local Bai3.pig
```

```
Input(s):
Successfully read 11770 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/hotel-re
view.csv"

Output(s):
Successfully stored 2 records in: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/KetQua_Bai3"

Counters:
Total records written : 2
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local243967829_0009 ->      job_local1109165004_0011,job_local347939181_0010,
job_local1109165004_0011        ->      job_local1993931565_0012,
job_local1993931565_0012        ->      job_local568572449_0014,
job_local568572449_0014 ->      job_local1656128800_0016,
job_local347939181_0010 ->      job_local536747876_0013,
job_local536747876_0013 ->      job_local1883818091_0015,
job_local1883818091_0015        ->      job_local1656128800_0016,
job_local1656128800_0016
```

```
 dy initialized!
 2026-04-10 02:20:49,113 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,115 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,117 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,120 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,122 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,124 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,131 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,135 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,137 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,145 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,150 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,153 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
 dy initialized!
 2026-04-10 02:20:49,155 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Su
 ccess!
 2026-04-10 02:20:49,199 [main] INFO  org.apache.pig.Main - Pig script completed in 24 seconds and 108 milliseconds (2410
 8 ms)
 thien592005@Thien-dz:/mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy$
```

Bài 4:

```
pig -x local Bai4.pig
```

```
Input(s):
Successfully read 131 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/stopwords.
txt"
Successfully read 11770 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/hotel-re
view.csv"

Output(s):
Successfully stored 8 records in: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/KetQua_Bai4_
Pos"
Successfully stored 8 records in: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/KetQua_Bai4_
Neg"

Counters:
Total records written : 16
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local132015807_0007 ->      job_local391068021_0008,
job_local391068021_0008 ->      job_local12542974_0009,job_local1935875729_0010,
job_local12542974_0009
job_local1935875729_0010
```

```
2026-04-10 02:23:54,395 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,413 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,422 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,442 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,446 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,452 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,462 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,471 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,477 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,492 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,496 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,500 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:23:54,513 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Su
ccess!
2026-04-10 02:23:54,668 [main] INFO  org.apache.pig.Main - Pig script completed in 40 seconds and 605 milliseconds (4060
5 ms)
thien592005@Thien-dz:/mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy$
```

Bài 5:

```
pig -x local Bai5.pig
```

```
vgReduceTime    MedianReducetime        Alias   Feature Outputs
job_local1215081673_0004        1       0       n/a     n/a     n/a     n/a     0       0       0       0       stopword
s       MAP_ONLY
job_local198337149_0006 1       1       n/a     n/a     n/a     n/a     n/a     n/a     n/a     n/a     group_by_cat,sor
ted,top5_related        GROUP_BY        file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/KetQua_
Bai5,
job_local407891610_0005 1       1       n/a     n/a     n/a     n/a     n/a     n/a     n/a     n/a     cleaned_data,dat
a_final,group_cat_word,joined_data,lower_data,raw_data,word_counts,words_flat   REPLICATED_JOIN,GROUP_BY,COMBINER

Input(s):
Successfully read 131 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/stopwords.
txt"
Successfully read 11770 records from: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/hotel-re
view.csv"

Output(s):
Successfully stored 8 records in: "file:///mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy/KetQua_Bai5"

Counters:
Total records written : 8
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1215081673_0004        ->      job_local407891610_0005,
job_local407891610_0005 ->      job_local198337149_0006,
job_local198337149_0006
```

```
Job DAG:
job_local1215081673_0004        ->      job_local407891610_0005,
job_local407891610_0005 ->      job_local198337149_0006,
job_local198337149_0006


2026-04-10 02:28:11,501 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,504 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,507 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,513 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,517 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,521 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,527 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,531 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,533 [main] WARN  org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system alrea
dy initialized!
2026-04-10 02:28:11,538 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Su
ccess!
2026-04-10 02:28:11,610 [main] INFO  org.apache.pig.Main - Pig script completed in -1 minutes, -41 seconds and -441 mill
iseconds (-101441 ms)
thien592005@Thien-dz:/mnt/c/Users/Admin/Documents/Uni/DS200/ThucHanh/BTTH2/Bai1 - Copy$
```